

基于 PaddleOCR-VL 的电路原理图 OCR 研究与实现

Circuit Schematic OCR via LoRA Fine-tuning of PaddleOCR-VL-0.9B

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<https://github.com/ZhangJ83/circuit-ocr-paddle>

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Code & Dataset: github.com/ZhangJ83/circuit-ocr-paddle

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Motivation: Why Circuit OCR?

Problem

- Circuit schematics contain dense technical text:
 - Component labels: R1, C2, U3, J4
 - Values: 10k Ω , 100nF, 3.3V
 - Pin numbers, net names, BOM tables
- Manual transcription is error-prone and slow
- Existing OCR tools (Tesseract, EasyOCR) fail on schematic fonts

Our Approach

- ① Fine-tune **PaddleOCR-VL-0.9B** — a Vision-Language Model
- ② Use **LoRA** for parameter-efficient training on consumer GPU (RTX 4060 8GB)
- ③ Design **anti-collapse training strategy** for small datasets
- ④ Evaluate with **multi-metric benchmark**

LoRA Configuration

- $r = 16$, $\alpha = 32$, dropout
- Target: Q/K/V/O proj + MLP layers
- **5.73M** trainable / 0.9B total (0.6%)
- Projector **frozen** (prevents modality collapse)

Training Details

- Optimizer: AdamW, weight_decay=0.1
- LR: LinearWarmup + CosineAnnealing
- Grad Accum = 4, Batch Size = 1
- Mixed precision: bfloat16
- Hardware: RTX 4060 8GB

Phase 1: Experiment Design

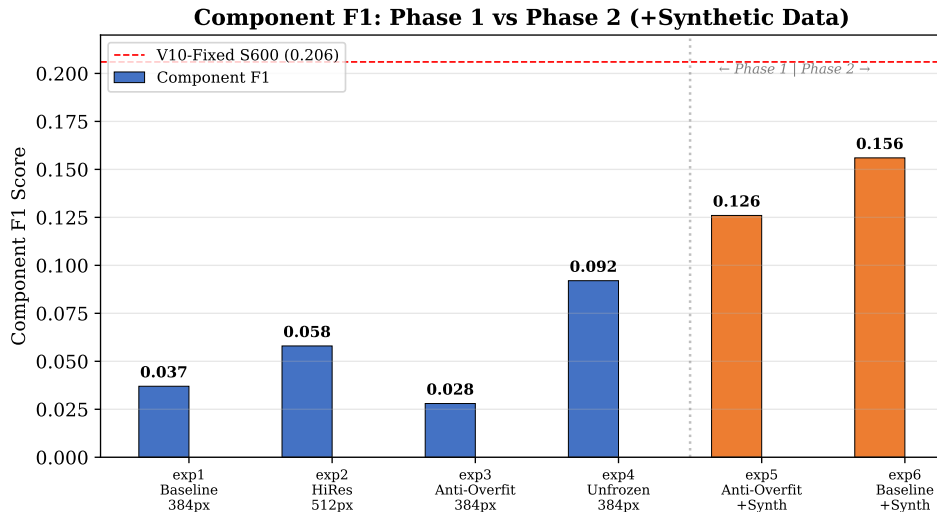
Experiment	Resolution	LR	Dropout	Epochs	Special
exp1 (Baseline)	384px	2×10^{-5}	0.05	2	—
exp2 (HiRes)	512px	2×10^{-5}	0.05	2	—
exp3 (Anti-Overfit)	384px	1×10^{-5}	0.10	3	Low LR + High DO
exp4 (Unfrozen)	384px	2×10^{-5}	0.05	2	Projector unfrozen

Motivation

- **exp3**: Hypothesis — lower LR + higher dropout prevents overfitting on 1200 samples
- **exp4**: Hypothesis — unfreezing projector allows better visual-text alignment
- **exp2**: Hypothesis — 512px captures fine text details (pin numbers, small labels)

Data: 1200 clean circuit schematics (1080 train + 120 val)

Phase 1: Results — Component F1



Phase 1: Modality Collapse Analysis

Collapse Pattern

- **S200**: “1,2,3,4,5,6,...” (pure counting)
- **S400**: “+3.3V, +3.3V, +3.3V...” (voltage repetition)
- **S600**: “R1, 10k, 12V...” (partial recovery)
- **exp3 only**: S800 showed “R1\n10k\nR2\n10k...”

Root Cause

- Small dataset (1200 samples) — model learns to **ignore image** and predict frequent patterns
- Training loss decreases but validation quality degrades — **loss-quality mismatch**
- Frozen projector may limit visual feature extraction

→ **Solution: Inject synthetic text-image pairs to force visual-text alignment**

Synthetic Data Strategy

What We Did

- 1 Generated **300 synthetic text images**
- 2 Content: component tables, pinouts, net labels, BOM entries
- 3 Rendered as clean document-style images (white background, black text)
- 4 Mixed with original 1200 circuit images at **20% ratio**
- 5 Total: **1500 training samples**

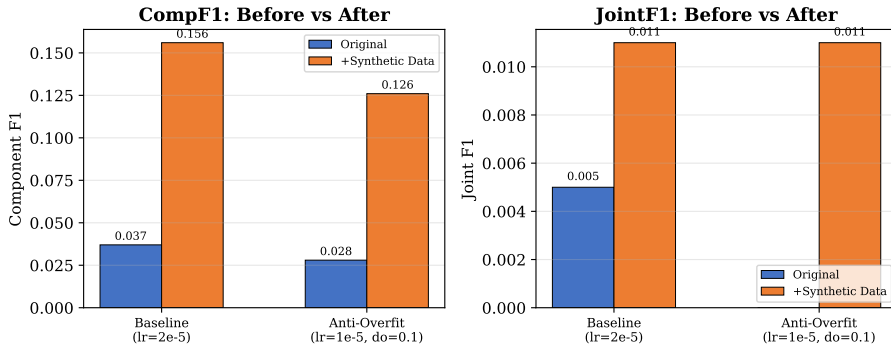
Why It Works

- Synthetic images have **perfect visual→text mapping**
- Model cannot “guess” the text — must read the image
- Forces the vision encoder to remain active
- Text content matches circuit domain (resistors, capacitors, voltages)
- Acts as a **regularizer** against counting collapse

1200 Circuit Images + 300 Synthetic Images = 1500 Mixed Dataset

Phase 2: Results

Synthetic Data Ablation: 300 Text-Image Pairs Mixed at 20%



Improvement

- **CompF1:** 0.037 $\rightarrow$ 0.156 (Baseline, $4.2\times$ improvement)
- **CompF1:** 0.028 $\rightarrow$ 0.126 (Anti-Overfit, $4.5\times$ improvement)
- **JointF1:** 0.005 $\rightarrow$ 0.011 ($2.2\times$ improvement)

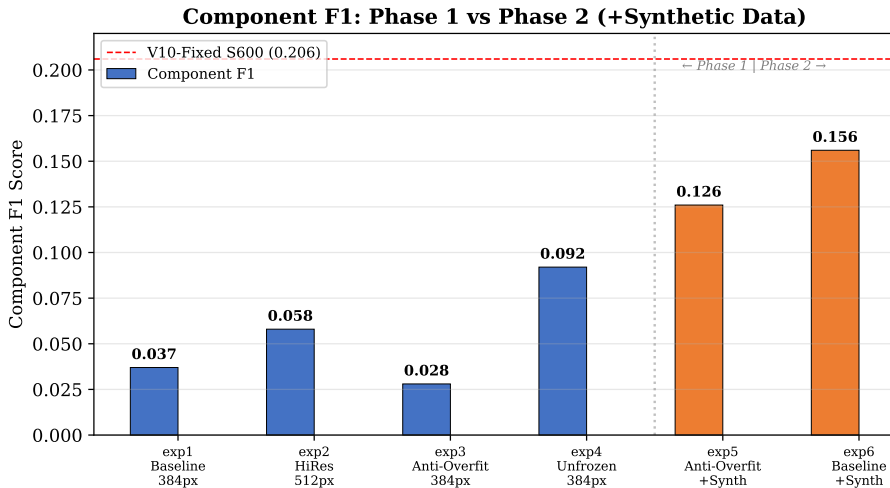
Training Quality Evolution



Key Insight

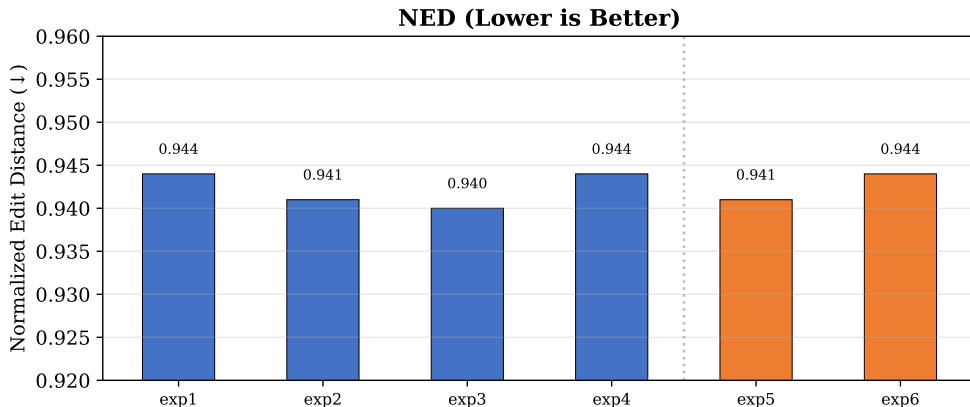
- Without synthetic data (exp3): gradual improvement, peaks at S800, still partial
- With synthetic data (exp5): **immediate high quality at S200**, stable through S800
- Synthetic data provides **strong initialization signal** for visual-text alignment

All Experiments: CompF1 & JointF1



Metric	Phase 1 Best	Phase 2 Best	vs V10-Fixed S600
CompF1	0.092 (exp4)	0.156 (exp6)	0.206 (3800 samples)

NED Comparison



Observation

NED remains consistently high (~ 0.94) across all experiments — even with improved CompF1 and JointF1, the edit distance is dominated by repetitive/collapsed predictions on unseen samples.

Key Findings

What Worked

- ① **Synthetic text data mixing** — single most effective intervention ($4\times$ CompF1 gain)
- ② **Anti-overfit config** (exp3/exp5): low LR + high dropout gave **stable** training
- ③ **LoRA + frozen projector** — prevents catastrophic modality collapse
- ④ **Multi-metric evaluation** — CompF1 + JointF1 + NED + RepRate captures different failure modes

What Didn't Work

- ① **512px high-res** (exp2) — no benefit, similar collapse
- ② **Unfrozen projector** (exp4) — higher CompF1 but still collapsed on most samples
- ③ **Loss-based checkpoint selection** — lowest loss $\neq$ best quality
- ④ **1200 samples insufficient** — even with synthetic data, $\text{joint_f1} \ll 0.05$ target

Core Insight: Synthetic data teaches visual-text alignment, but real schematic diversity is needed for generalization.

Limitations & Future Work

Current Limitations

- JointF1 (0.011) far below usable threshold (0.05)
- NED (~ 0.94) indicates poor per-token accuracy
- 1200 training samples insufficient for generalization
- Real-camera photos not yet integrated
- Validation set (3 samples) too small for reliable monitoring

Future Directions

- ① **Scale up data:** collect 3000+ diverse real schematics
- ② **Real-camera data:** add photographed circuit boards/prints
- ③ **More synthetic variety:** fonts, sizes, layouts, noise
- ④ **Larger LoRA rank:** try $r = 32$ or $r = 64$
- ⑤ **Two-stage training:** synthetic pre-train $\rightarrow$ real fine-tune
- ⑥ **Better evaluation:** 50+ sample validation set

Summary

Project Status

- Built complete **training + evaluation pipeline** for circuit schematic OCR
- Achieved **CompF1 = 0.156** with synthetic data mixing (Phase 2)
- Demonstrated **anti-collapse training strategy** via synthetic text-image injection
- **JointF1 = 0.011** — model beginning to learn (refdes, value) pairs but not yet usable

Next Milestone

- ① Collect **larger dataset** (3000+ real schematics)
- ② Integrate **real-camera photos** for robustness
- ③ Target: **JointF1 $\geq$ 0.05, CompF1 $\geq$ 0.25**

github.com/ZhangJ83/circuit-ocr-paddle

Full code, dataset, training scripts, and experiment logs available.

Thank You!

谢谢!

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PaddleOCR-VL-0.9B + LoRA Fine-tuning
Circuit Schematic OCR